

A Cyclic Polarized Isogeny on a Simple Abelian Threefold

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Abstract

We construct a simple complex abelian threefold A and a polarized endomorphism $f : A \rightarrow A$ such that f is not incompressible, in the sense of essential dimension. More precisely, f is 4-polarized, $\deg f = 64$, and $\ker(f) \simeq \mathbb{Z}/64\mathbb{Z}$. Since a cyclic cover over $\mathbb{C}$ has essential dimension at most one, while $\dim A = 3$, this gives a concrete negative answer to the question whether every polarized endomorphism of a simple abelian variety is incompressible.

1 Definitions and main result

Throughout the paper the base field is $\mathbb{C}$. If $\pi : Y \dashrightarrow X$ is a generically finite dominant rational map of irreducible varieties, its essential dimension is denoted by $\mathrm{ed}(\pi)$. We call π *incompressible* if

$$\mathrm{ed}(\pi) = \dim X = \dim Y.$$

Equivalently, π cannot be birationally obtained by base change from a generically finite map of smaller-dimensional varieties.

We shall use the following standard fact. If π is a finite abelian Galois cover with Galois group G , then

$$\mathrm{ed}(\pi) \leq \mathrm{rank}(G),$$

where $\mathrm{rank}(G)$ is the minimum number of generators of G . In particular, a cyclic cover over $\mathbb{C}$ has essential dimension at most one. Indeed, by Kummer theory, a cyclic degree- n extension of a field containing the n -th roots of unity is obtained from the one-dimensional cover

$$\mathbb{G}_m \longrightarrow \mathbb{G}_m, \quad t \longmapsto t^n.$$

Theorem 1. *There exists a simple abelian threefold A over $\mathbb{C}$ and a polarized endomorphism $f : A \rightarrow A$ such that f is not incompressible. More precisely, there is a 4-polarized isogeny f with*

$$\ker(f) \simeq \mathbb{Z}/64\mathbb{Z}, \quad \mathrm{ed}(f) = 1 < 3 = \dim A.$$

2 The CM abelian threefold

Let

$$\zeta = \zeta_7 = e^{2\pi i/7}, \quad K = \mathbb{Q}(\zeta).$$

Then K is a CM field of degree 6. Let

$$\sigma_j : K \hookrightarrow \mathbb{C}, \quad \sigma_j(\zeta) = \zeta^j, \quad j = 1, 2, 3,$$

and choose the CM type

$$\Phi = \{\sigma_1, \sigma_2, \sigma_3\}.$$

Put

$$\alpha = 2\zeta, \quad R = \mathbb{Z}[\alpha] = \mathbb{Z}[2\zeta].$$

Via the CM type Φ , regard R as a lattice in $\mathbb{C}^3$ by

$$x \longmapsto \Phi(x) = (\sigma_1(x), \sigma_2(x), \sigma_3(x)).$$

Define the complex torus

$$A = \mathbb{C}^3 / \Phi(R).$$

2.1 Projectivity

We first show that A is an abelian variety. Let

$$\delta = \zeta^{-1} - \zeta.$$

Then $\bar{\delta} = -\delta$. For $x, y \in R$, define

$$E(x, y) = \text{Tr}_{K/\mathbb{Q}}(\delta x \bar{y}).$$

Since $\delta, x, \bar{y}$ are algebraic integers, $E(x, y) \in \mathbb{Z}$. Moreover E is alternating, because

$$E(y, x) = \text{Tr}_{K/\mathbb{Q}}(\delta y \bar{x}) = \text{Tr}_{K/\mathbb{Q}}(\overline{\delta y \bar{x}}) = \text{Tr}_{K/\mathbb{Q}}(-\delta x \bar{y}) = -E(x, y).$$

For $j = 1, 2, 3$ we have

$$\text{Im } \sigma_j(\delta) = -2 \sin(2\pi j/7) < 0.$$

Thus E is the imaginary part of the positive definite Hermitian form

$$H(z, w) = \sum_{j=1}^3 c_j z_j \bar{w}_j, \quad c_j = -2 \text{Im } \sigma_j(\delta) > 0.$$

Indeed, for $x, y \in K$ one checks that

$$E(x, y) = \sum_{j=1}^3 c_j \text{Im}(\sigma_j(x) \overline{\sigma_j(y)}).$$

Hence E is a Riemann form on the torus A , and A is projective. Therefore A is an abelian variety.

We shall use a slightly scaled line bundle. Since E is integral, the form $E_0 = 2E$ is even integral. By the Appell–Humbert theorem, the Riemann form E_0 together with the trivial semicharacter defines an ample line bundle L on A .

2.2 Simplicity

Lemma 1. *The abelian variety A is simple.*

Proof. The lattice $R = \mathbb{Z}[2\zeta]$ has finite index in the full ring of integers $\mathcal{O}_K = \mathbb{Z}[\zeta]$. Hence A is isogenous to

$$A_0 = \mathbb{C}^3 / \Phi(\mathcal{O}_K).$$

It is therefore enough to prove that A_0 is simple.

The Galois group of $K/\mathbb{Q}$ is

$$\text{Gal}(K/\mathbb{Q}) \simeq (\mathbb{Z}/7\mathbb{Z})^\times.$$

The only proper CM subfield of K is the quadratic subfield $\mathbb{Q}(\sqrt{-7})$. The subgroup fixing it is

$$H = \{1, 2, 4\} \subset (\mathbb{Z}/7\mathbb{Z})^\times.$$

Therefore the CM types of K induced from this quadratic CM subfield are precisely

$$\{1, 2, 4\} \quad \text{and} \quad \{3, 5, 6\}.$$

Our CM type is

$$\Phi = \{1, 2, 3\},$$

which is neither of these. Thus Φ is primitive. By the standard criterion for complex multiplication, a CM abelian variety with primitive CM type is simple. Hence A_0 is simple, and since simplicity is invariant under isogeny, A is simple. $\square$

3 The polarized endomorphism

Multiplication by

$$\alpha = 2\zeta$$

preserves the lattice $R = \mathbb{Z}[\alpha]$. It therefore defines an endomorphism

$$f : A \longrightarrow A, \quad z \longmapsto \alpha z,$$

where on the universal cover $\mathbb{C}^3$ this means diagonal multiplication by

$$(\sigma_1(\alpha), \sigma_2(\alpha), \sigma_3(\alpha)).$$

Since $\alpha \neq 0$, this endomorphism is an isogeny.

Lemma 2. *The endomorphism f is 4-polarized with respect to the ample line bundle L constructed above. In particular,*

$$\deg(f) = 4^3 = 64.$$

Proof. For $x, y \in R$,

$$\begin{aligned} E(\alpha x, \alpha y) &= \text{Tr}_{K/\mathbb{Q}}(\delta \alpha x \overline{\alpha y}) \\ &= \text{Tr}_{K/\mathbb{Q}}(\delta \alpha \overline{\alpha} x \overline{y}). \end{aligned}$$

But

$$\alpha\bar{\alpha} = (2\zeta)(2\zeta^{-1}) = 4.$$

Therefore

$$E(\alpha x, \alpha y) = 4E(x, y), \quad E_0(\alpha x, \alpha y) = 4E_0(x, y).$$

The line bundle L was defined by the Riemann form E_0 and the trivial semicharacter. Pulling back by f multiplies the Riemann form by 4, and the trivial semicharacter remains trivial. Thus, by the Appell–Humbert theorem,

$$f^*L \simeq L^{\otimes 4}.$$

So f is 4-polarized. Since A has dimension 3, a 4-polarized endomorphism has degree $4^3 = 64$. $\square$

Lemma 3. *The kernel of f is cyclic of order 64:*

$$\ker(f) \simeq \mathbb{Z}/64\mathbb{Z}.$$

Proof. The kernel is

$$\ker(f) \simeq \alpha^{-1}R/R.$$

Multiplication by α identifies this group with $R/\alpha R$.

The minimal polynomial of $\alpha = 2\zeta$ is

$$\begin{aligned} P(t) &= 2^6 \Phi_7(t/2) \\ &= t^6 + 2t^5 + 4t^4 + 8t^3 + 16t^2 + 32t + 64. \end{aligned}$$

Hence

$$R = \mathbb{Z}[\alpha] \simeq \mathbb{Z}[t]/(P(t)).$$

Modulo the ideal generated by α , equivalently by t , we get

$$\begin{aligned} R/\alpha R &\simeq \mathbb{Z}[t]/(P(t), t) \\ &\simeq \mathbb{Z}/(P(0)) \\ &\simeq \mathbb{Z}/64\mathbb{Z}. \end{aligned}$$

Thus $\ker(f)$ is cyclic of order 64. $\square$

4 Failure of incompressibility

The isogeny $f : A \rightarrow A$ is finite etale and Galois; its Galois group is the translation group by its kernel:

$$\mathrm{Gal}(f) \simeq \ker(f) \simeq \mathbb{Z}/64\mathbb{Z}.$$

Since this group is cyclic, the essential dimension of the cover is at most one:

$$\mathrm{ed}(f) \leq 1.$$

On the other hand f has degree $64 > 1$, so it is not birational. A nontrivial finite cover over an algebraically closed base field cannot have essential dimension zero. Hence

$$\mathrm{ed}(f) = 1.$$

But

$$\dim A = 3.$$

Therefore

$$\mathrm{ed}(f) = 1 < 3 = \dim A,$$

and f is not incompressible. This proves Theorem 1.

5 Where the hypotheses are used

- (1) The fact that A is an abelian variety is obtained from the explicit Riemann form

$$E(x, y) = \mathrm{Tr}_{K/\mathbb{Q}}((\zeta^{-1} - \zeta)x\bar{y}).$$

This is the projectivity input.

- (2) The simplicity of A comes from the primitive CM type

$$\Phi = \{1, 2, 3\} \subset (\mathbb{Z}/7\mathbb{Z})^\times.$$

The only proper CM subfield of $\mathbb{Q}(\zeta_7)$ is $\mathbb{Q}(\sqrt{-7})$, and the types induced from it are $\{1, 2, 4\}$ and $\{3, 5, 6\}$, not $\{1, 2, 3\}$.

- (3) The polarizedness of f uses the identity

$$(2\zeta)(2\zeta^{-1}) = 4.$$

It implies that multiplication by 2ζ multiplies the chosen Riemann form by 4, hence $f^*L \simeq L^{\otimes 4}$.

- (4) The failure of incompressibility uses the non-maximal order

$$R = \mathbb{Z}[2\zeta]$$

rather than the full ring of integers. This choice makes

$$R/(2\zeta)R \simeq \mathbb{Z}/64\mathbb{Z},$$

so the Galois group of the isogeny is cyclic. A cyclic cover has essential dimension at most one, which is strictly smaller than $\dim A = 3$.

Remark 1. The example is constructed analytically over $\mathbb{C}$. Since it is a CM abelian variety with CM endomorphism 2ζ , the usual theory of complex multiplication also gives models over $\overline{\mathbb{Q}}$ after replacing it by an isomorphic algebraic model.

References

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